SMART MEALS FOR PEOPLE WITH DIABETES

A COMMUNITY SERVICE PROJECT REPORT

Submitted to

Jawaharlal Nehru Technological University Kakinada

In partial fulfillment for the award of the degree of

Bachelor of Technology

In

CSE – DATA SCIENCE

Submitted by

BOMMALI DILEEP (24KN1A4425)

Under the esteemed guidance of

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DEPARTMENT OF CSE – DATA SCIENCE

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Pothavarappadu (V), (Via) Nunna, Agiripalli (M), Krishna Dist, PIN: 521212, A.P, India.

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Certificate

This is to certify that the Community Service Project report entitled **"SMART MEALS FOR PEOPLE WITH DIABETES"** is a bonafide work carried out by **BOMMALI DILEEP (24KN1A4425)** in partial fulfillment for the award of degree of **Bachelor of Technology in CSE – DATA SCIENCE** of **Jawaharlal Nehru Technological University Kakinada**, during the academic year **2026-27**.

Project Guide

Ms. T. Navya Chandrika
Assistant Professor
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Head of the Department

Dr. CH. V. Murali Krishna
Professor & HOD
Dept. of CSE – Data Science

DECLARATION

I hereby declare that the Community Service Project report titled **"SMART MEALS FOR PEOPLE WITH DIABETES"** is a bonafide work carried out in the **Department of CSE-DATA SCIENCE**, NRI Institute of Technology, Agiripalli, Vijayawada, during the academic year **2026-2027**, in partial fulfillment for the award of the degree of **Bachelor of Technology** by **JNTU Kakinada**.

I further declare that this dissertation has not been submitted elsewhere for any Degree or Diploma.

Place: Agiripalli
Date:

BOMMALI DILEEP
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ABSTRACT

Diabetes mellitus represents one of the most critical and fast-growing public health challenges of the 21st century, with India often recognized as the diabetes capital of the world. Effective clinical control and sustained lifestyle management of diabetes rely fundamentally on medical nutrition therapy, glycemic index comprehension, strict portion discipline, and routine blood glucose monitoring. The Community Service Project (CSP) titled **"SMART MEALS FOR PEOPLE WITH DIABETES"** was executed over an intensive eight-week duration (May 3, 2026 to June 27, 2026) focusing on diabetic outpatients, resident caregivers, and healthcare visitors at Government Hospital, Gunadala, Kanuru, Vijayawada, under the academic supervision of **Ms. T. Navya Chandrika**, Assistant Professor, Department of CSE – Data Science, NRI Institute of Technology.

The primary objective of the initiative was to evaluate grassroots community nutritional literacy, identify deep-rooted cultural dietary misconceptions (including irregular meal spacing, excessive reliance on refined polished white rice, and fear of healthy fats), and formulate practical, culturally compatible dietary frameworks. Through bilingual structured socio-economic surveys, one-on-one dietary consultations, and demographic analysis, quantitative and qualitative data was gathered regarding carbohydrate intake habits, postprandial glucose tracking awareness, and financial constraints associated with obtaining balanced nutrition.

To bridge the gap between medical prescriptions and daily household cooking, the project implemented the evidence-based **"Smart Plate Method"**—advocating half-plate non-starchy fibrous vegetables, quarter-plate complex carbohydrates (unpolished millets, brown rice), and quarter-plate plant/dairy proteins (sprouts, pulses, curd). A memorable three-step dietary heuristic—**Recognise, Measure, and Pair**—was disseminated through visual portion charts and interactive demonstrations. Additionally, indigenous low-cost superfoods such as fenugreek seeds, bitter gourd, and cinnamon were integrated into affordable daily meal plans. The project achieved measurable enhancements in patient portion discipline, self-monitoring adherence, and community health empowerment, underscoring that structured grassroots nutritional counselling is a vital, cost-effective preventative strategy against severe diabetic complications.

Keywords: Diabetes Nutrition, Smart Meals, Glycemic Index, Plate Method, Community Health Education, Dietary Intervention, Preventative Healthcare, Rural Health.